How Does Data Science Impact the Semantic Web?

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BIOMEDICAL ENGINEERING

Disclaimer – A Broad But Shallow Discussion

- Not really sure what the semantic web is anymore
- At this point I can't give you a technical perspective
- Deeply engaged in preparing one academic institution for a very different data driven future

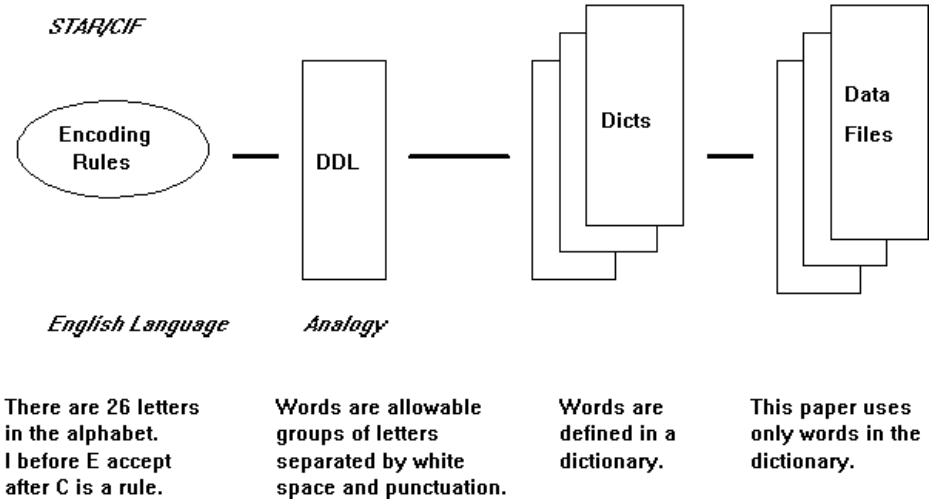




Biased by Lessons Learned a Long Time Ago ….

mmCIF - Extract from the Dictionary

```
save__atom_site.Cartn_x
   _item_description.description
;          The x atom site coordinate in angstroms specified according to
           a set of orthogonal Cartesian axes related to the cell axes as
           specified by the description given in
           _atom_sites.Cartn_transform_axes.
;
   _item.name                '_atom_site.Cartn_x'
   _item.category_id           atom_site
   _item.mandatory_code        no
   _item_aliases.alias_name    '_atom_site_Cartn_x'
   _item_aliases.dictionary    cifdic.c94
   _item_aliases.version       2.0
    loop_
   _item_dependent.dependent_name
                        '_atom_site.Cartn_y'
                        '_atom_site.Cartn_z'
   _item_related.related_name  '_atom_site.Cartn_x_esd'
   _item_related.function_code   associated_esd
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   _item_type.code             float
   _item_type_conditions.code    esd
   _item_units.code            angstroms
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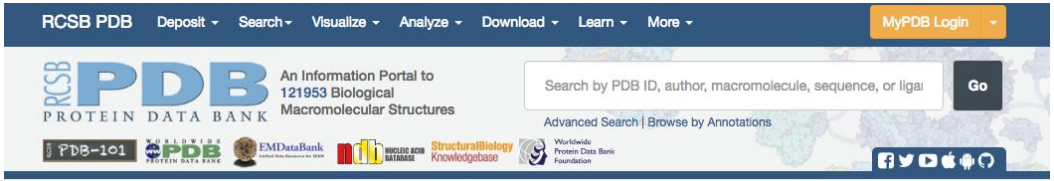


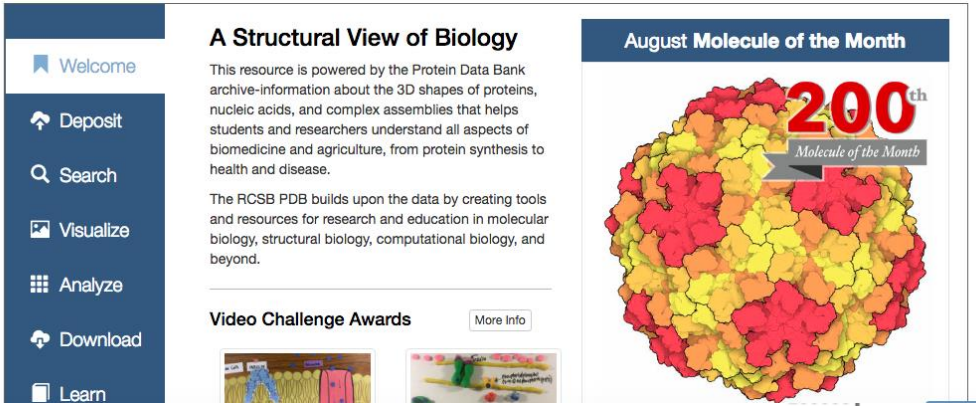
Bourne et al. 1997 *Meth. Enz*. 277 571-590

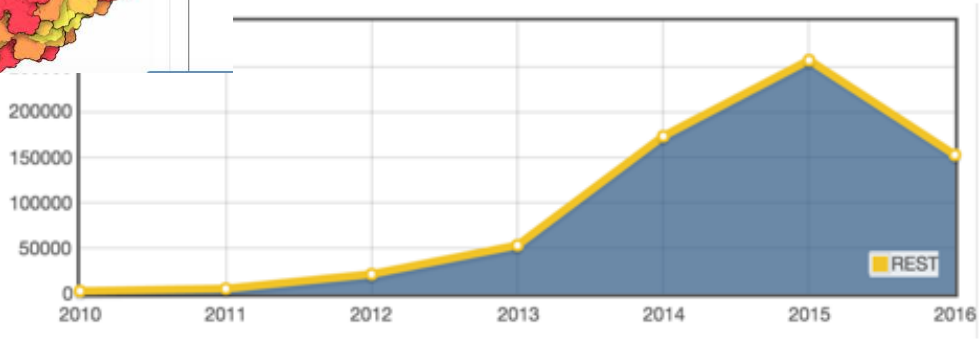
Lessons Learned a Long Time Ago

- Science is what happens when you are writing formal definitions
- Define the intended audience and focus on catering to them
- Keep it simple
- Back up that simplicity with software
- It can take many years for the effort to pay off

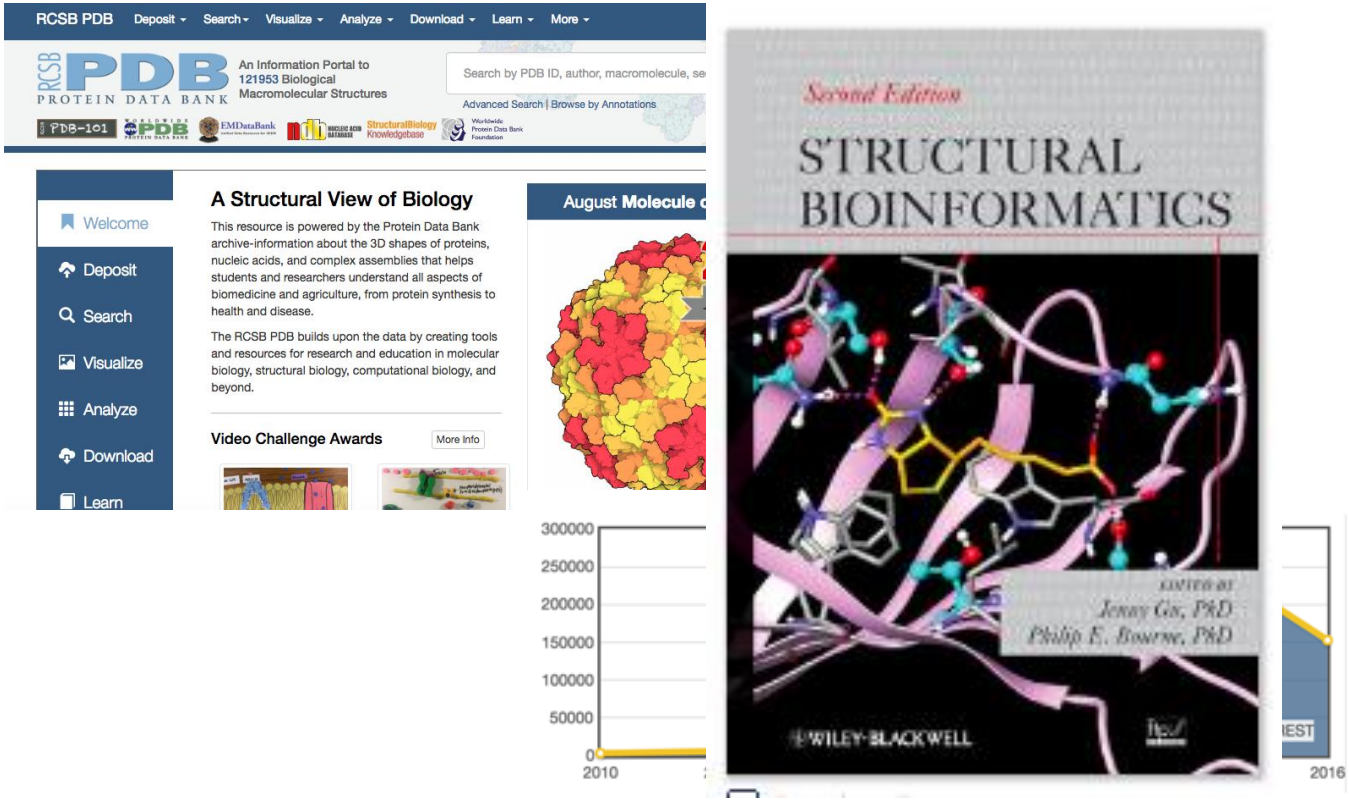
RCSB Protein Data Bank 1999-2014







RCSB Protein Data Bank 1999-2014



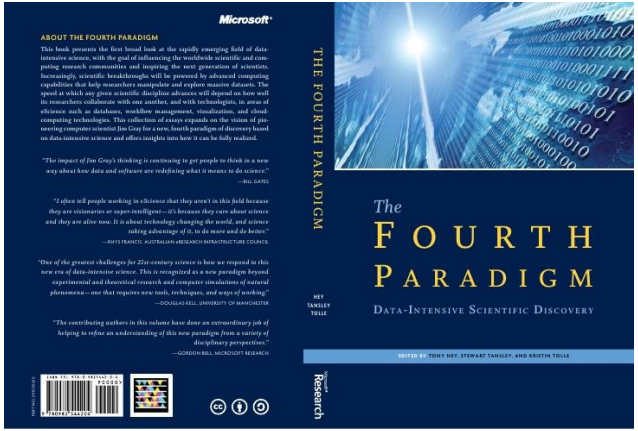
With that backdrop, lets return to our original question ….

How Does Data Science Impact the Semantic Web?

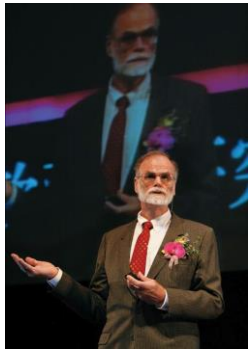
How Does Data Science Impact the Semantic Web….

The short answer {in my opinion} is profoundly …

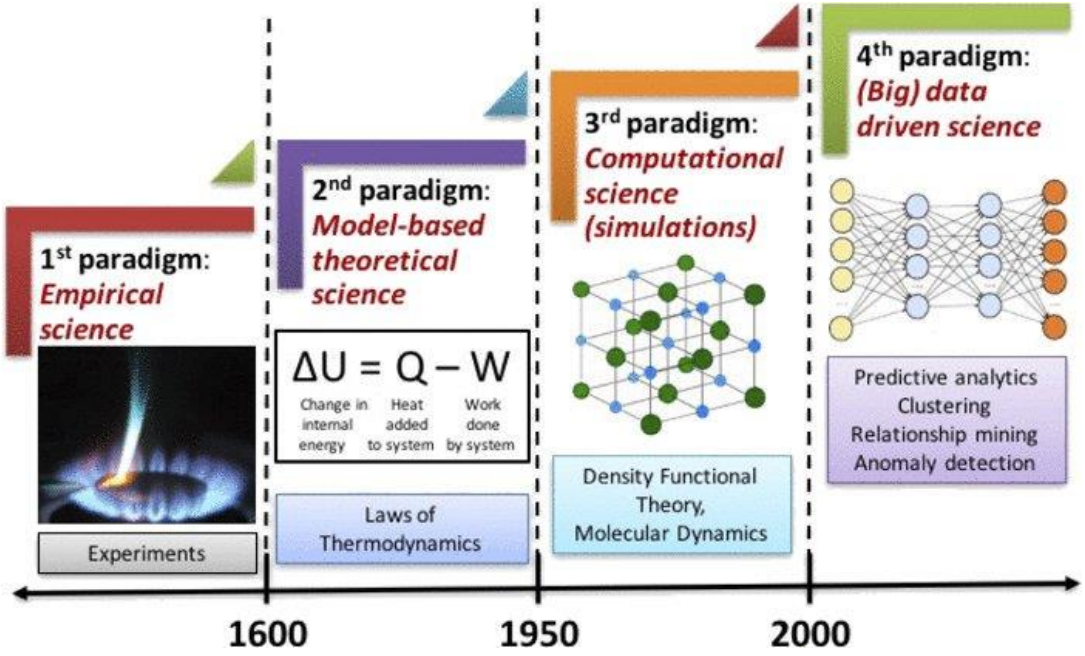
by virtue that data science is poised to impact everything



https://www.microsoft.com/en-us/research/wp-content/uploads/2009/10/Fourth_Paradigm.pdf

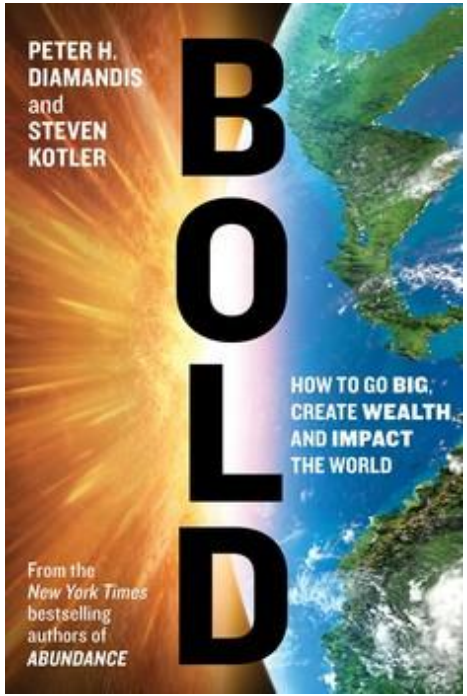


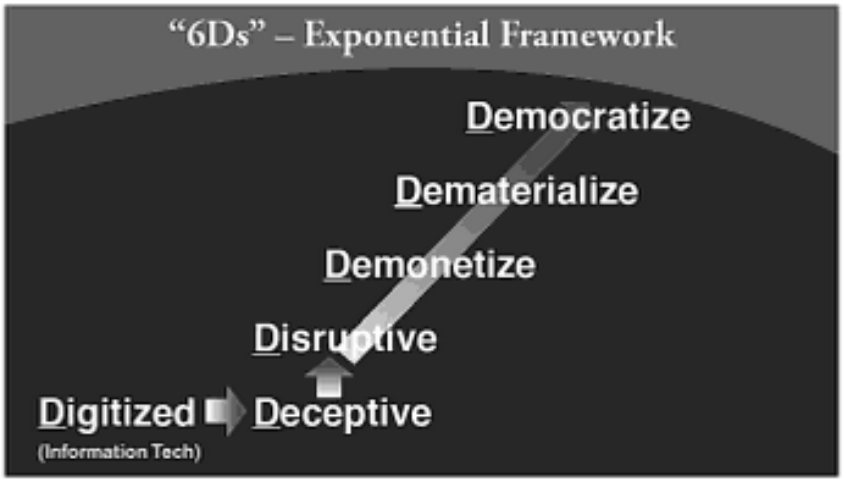
https://en.wikipedia.org/wiki/Jim_Gray_(computer_scientist)



https://twitter.com/aip_publishing/status/856825353645559808

How Will Science Change?

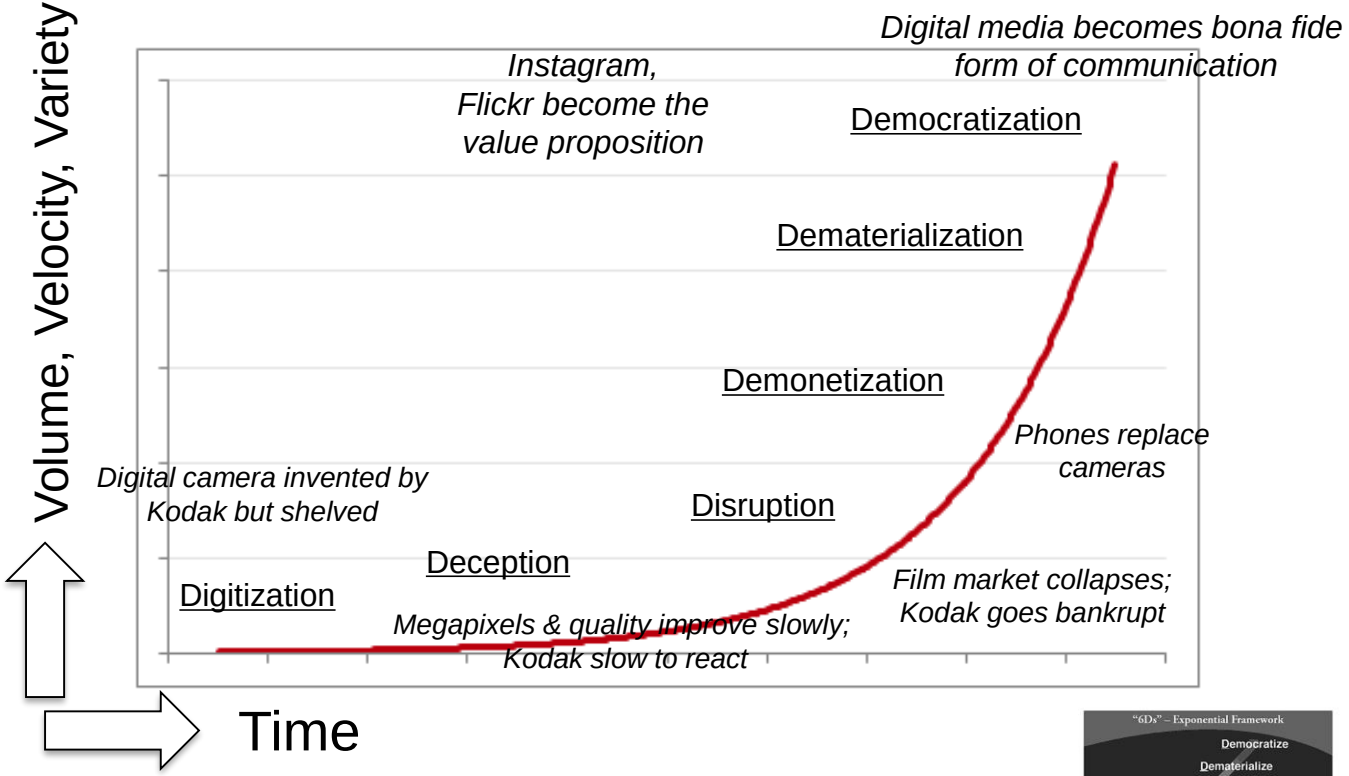




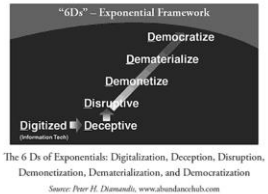
The 6 Ds of Exponentials: Digitalization, Deception, Disruption, Demonetization, Dematerialization, and Democratization

Source: Peter H. Diamandis, www.abundancehub.com

Example - Photography



From a presentation to the Advisory Board to the NIH Director



The 6 Ds of Exponentials: Digitalization, Deception, Disruption, Demonetization, Dematerialization, and Democratization

Source: Peter H. Diamandis, www.abundancehub.com

To build on this notion, we need working definition of data science …

It is the unexpected re-use of information which is the value added by the web

Tim Berners-Lee

https://www.forbes.com/sites/gilpress/2013/05/28/a-very-short-history-of-data-science/#116a5a2d55cf

To build on this notion we need working definition of data science …

It is the unexpected re-use of information which is the value added by the web and subsequent analysis of that information for societal benefit

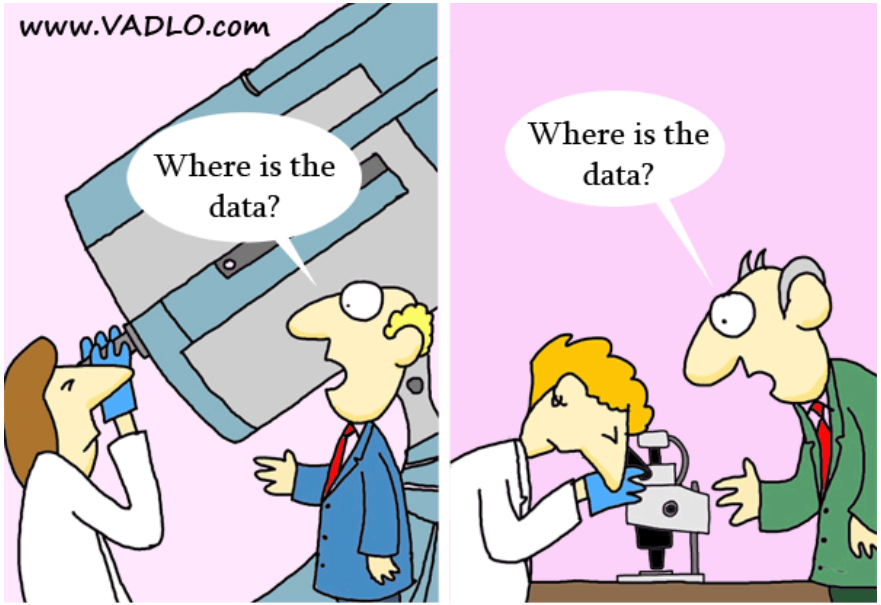
Tim Berners-Lee / Phil Bourne

https://www.forbes.com/sites/gilpress/2013/05/28/a-very-short-history-of-data-science/#116a5a2d55cf

To date, data science is too frequently the unexpected reuse of information without the {semantic} web!

Witness the tale of the trauma surgeon …

Data science is like the Internet… If I asked you to define it you would all say something different, yet you use it every day…



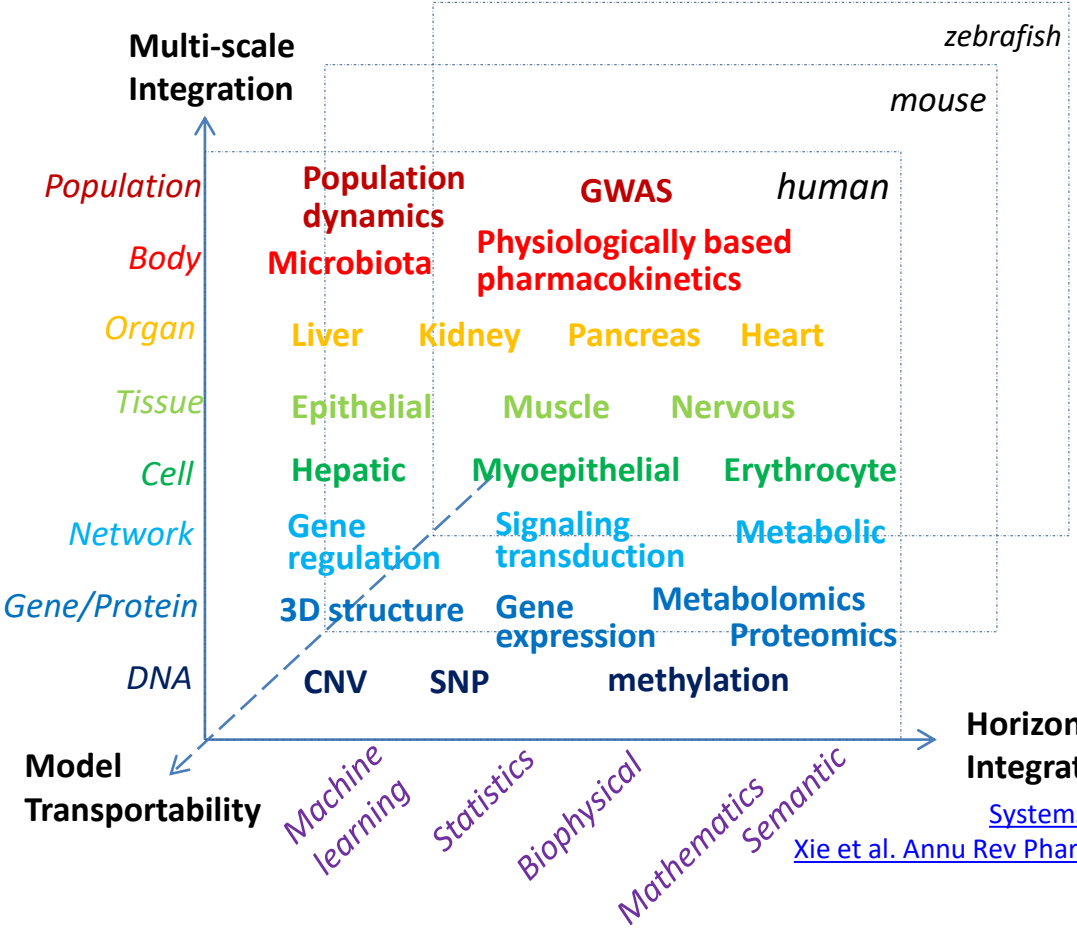
Grand Unification of Sciences

http://vadlo.com/cartoons.php?id=357

So What Do I Mean by Data Science?

- Use of the ever increasing amount of open, complex, diverse digital data
- Finding ways to ask and then answer relevant questions by combining such diverse data sets
- Arriving at statistically significant conclusions not otherwise obtainable
- Sharing such findings in a useful way
- Translating such findings into actions that improve the human condition

Open, complex, diverse digital data



Systems Pharmacology

Xie et al. Annu Rev Pharmacol Toxicol. 2017 57:245-262

Why Now?
Machine learning has been around for over 20 years

- Amount of data available for training
- Open source - R and Python
- Advances in computing (e.g., GPU's) allow for deeper neural nets (deep learning)
- Algorithmic efficiency gains (e.g., in back propagation)
- Success promotes further research
- Commercialization

Pastur-Romay et al. 2016 doi:10.3390/ijms17081313

Why Now? – Cost vs Use {Apologies} A US Centric View

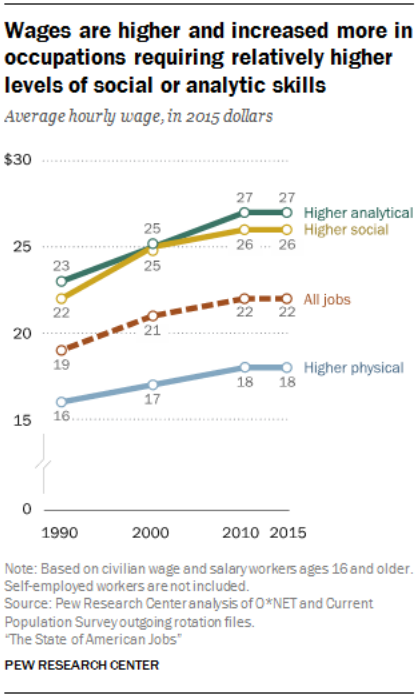
- Big Data
  - Total data from NIH-funded research back in 2016 estimated at 650 PB*
  - 20 PB of that is in NCBI/NLM (3%) and it is expected to grow by 10 PB in 2016
- Dark Data
  - Only 12% of data described in published papers is in recognized archives – 88% is dark data^
- Cost
  - 2007-2014: NIH spent ~$1.2Bn extramurally on maintaining data archives

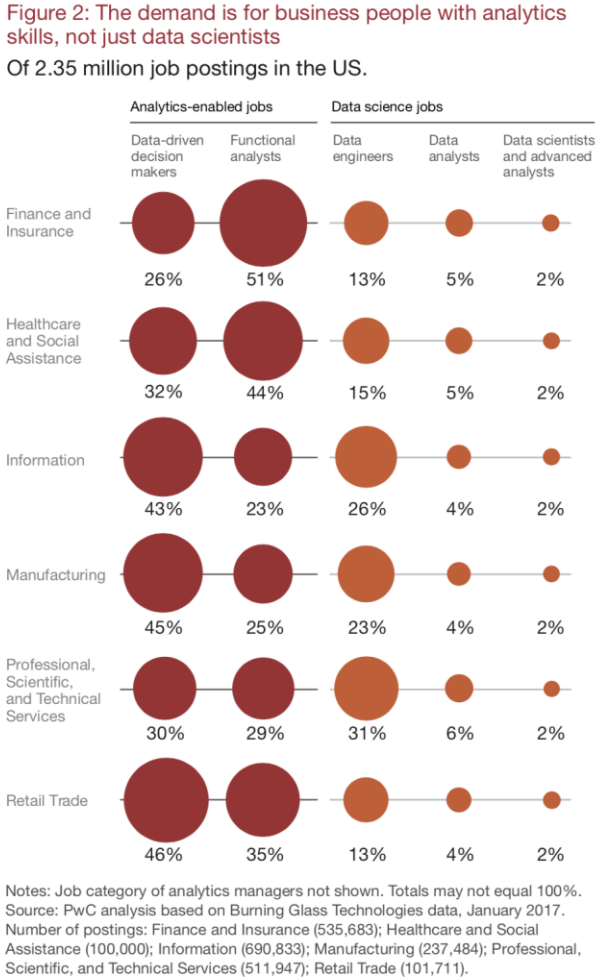
* In 2012 Library of Congress was 3 PB

^ http://www.ncbi.nlm.nih.gov/pubmed/26207759

Why Now? – Training

{More Apologies}

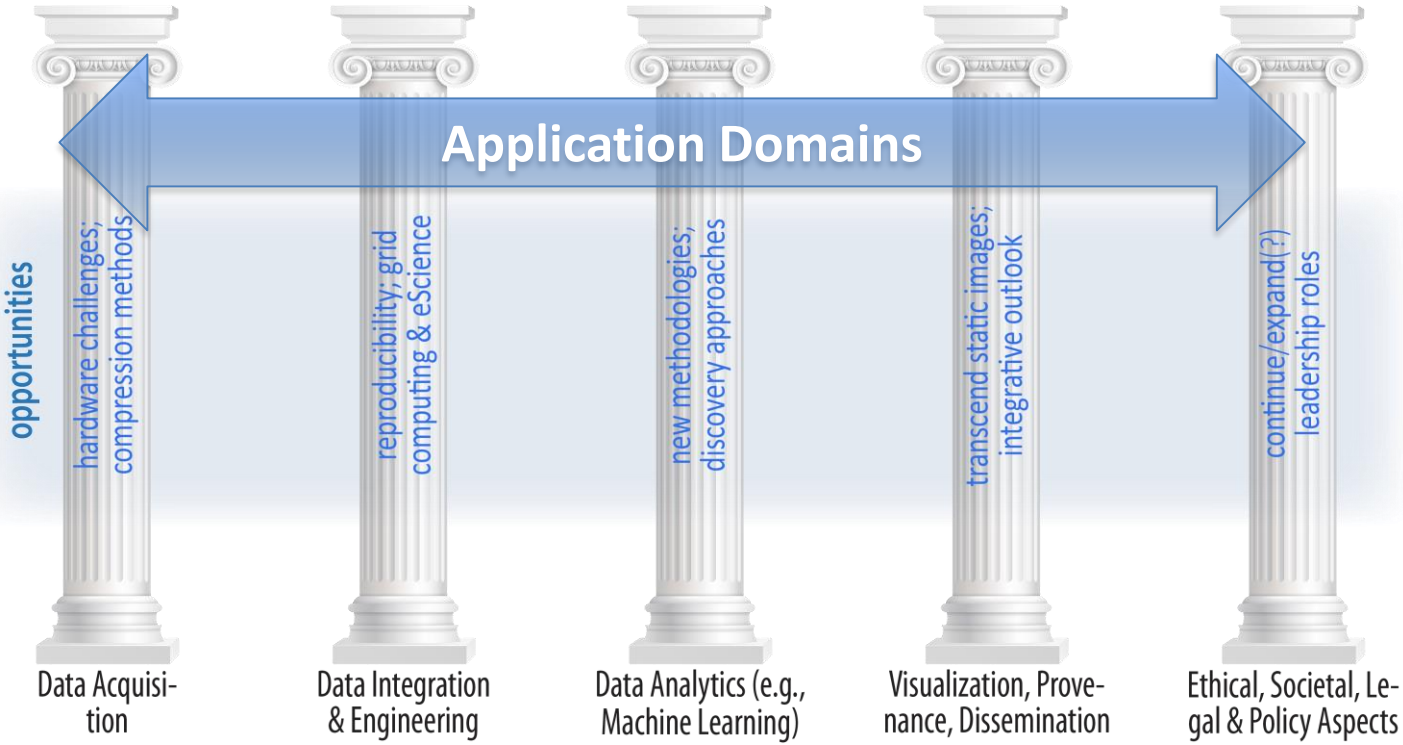




But here is the thing…

None of our current training programs, notably a MS in Data Science, cover the semantic web per se

The Pillars of Data Science



Lets briefly focus on those five pillars in the context of one area of biomedical informatics – structural bioinformatics

What kinds of interchange should be taking place between this field and data science?

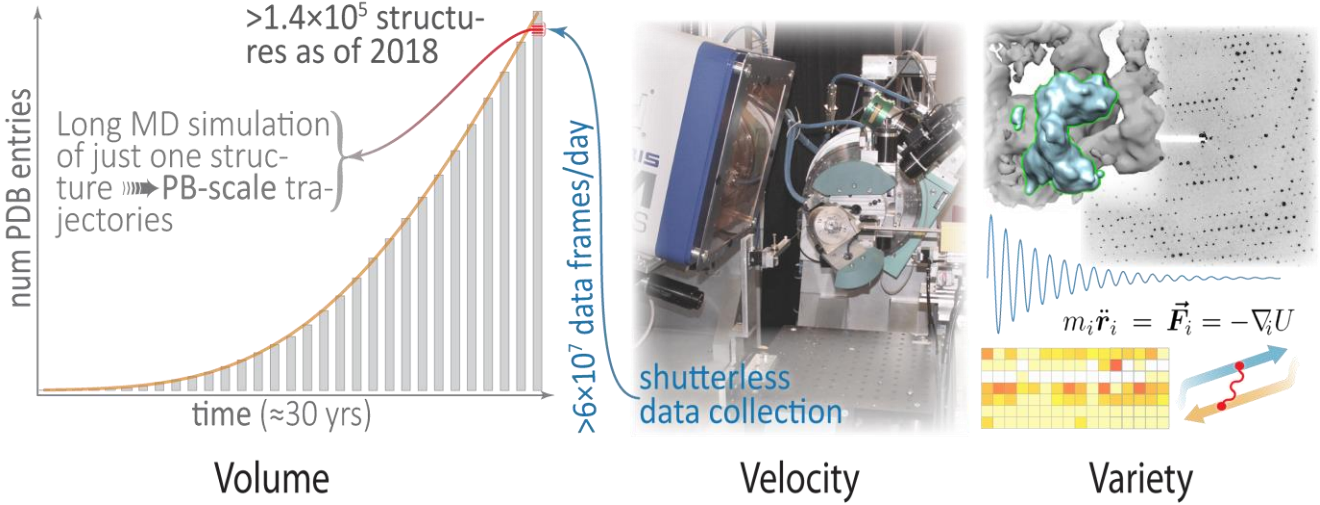
Mura et al. 2018 Curr Opin Struct Biol. 52:95-102



Data Acquisition

- Persistence of raw data not clear
- Some level of consistency across instrument manufacturers
- Lessons in community/society drive



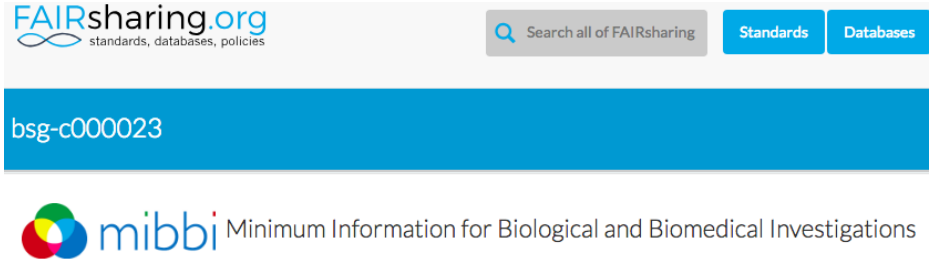


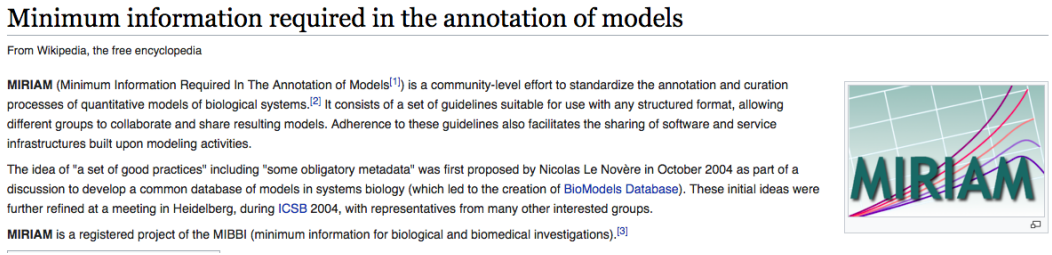
Mura et al. 2018 Curr Opin Struct Biol. 52:95-102



Data Integration and Engineering

- URI's no - stooped in tradition
- Ontologies – somewhat
- Linked data - somewhat



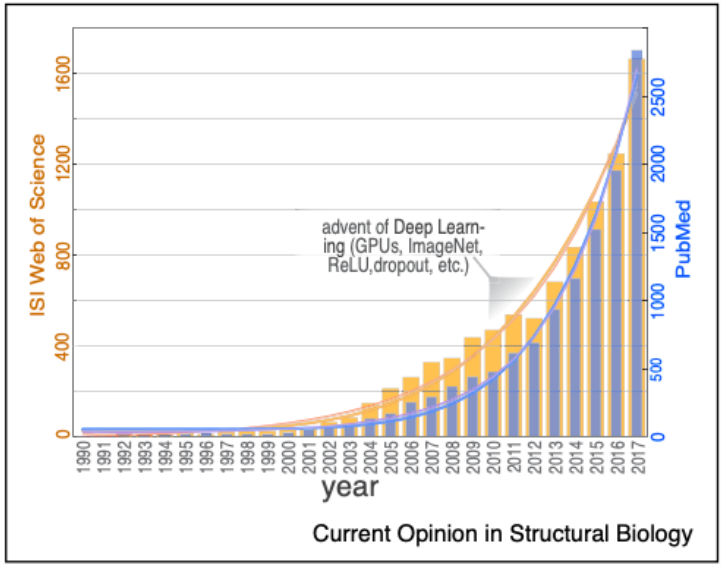


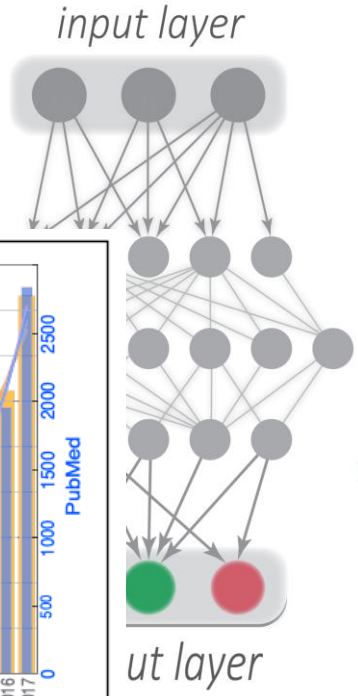
Years of experience to convey

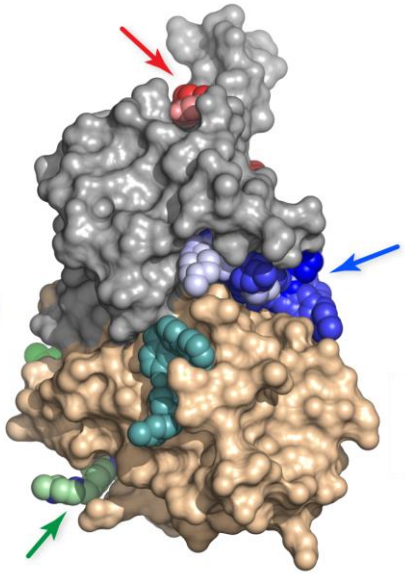


Data Analytics

- SVM's
- Random forest
- Neural nets
- Deep learning
- ??







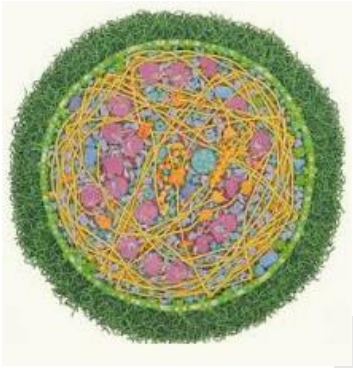
Opportunity to learn from many domains

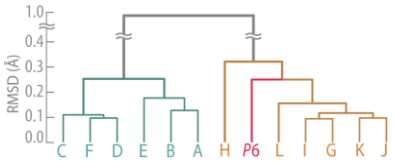


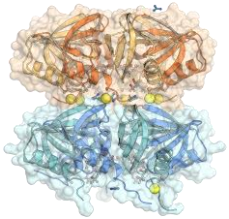


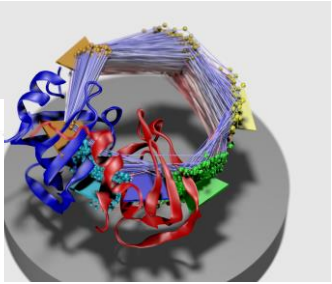
Visualization & Dissemination

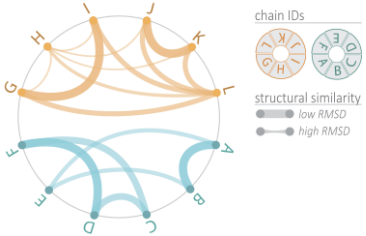
- Avoid the curse of the ribbon
- Think sonics
- Look to video games

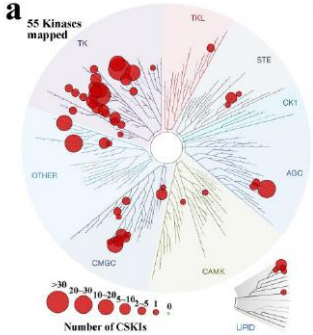


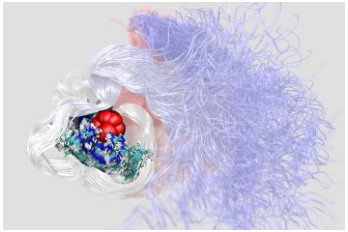










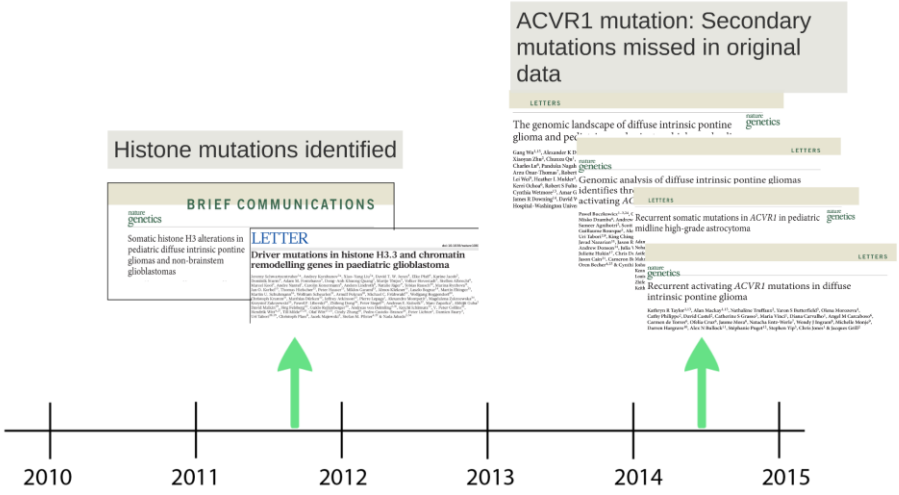




Ethics, Law & Policy – Data Sharing for Reuse

- Landmark studies identify histone mutations as recurrent driver mutations in DIPG ~2012
- Almost 3 years later, in largely the same datasets, but partially expanded, the same two groups and 2 others identify ACVR1 mutations as a secondary, co-occurring mutation

Diffuse Intrinsic Pontine Glioma (DIDG)



From Adam Resnick



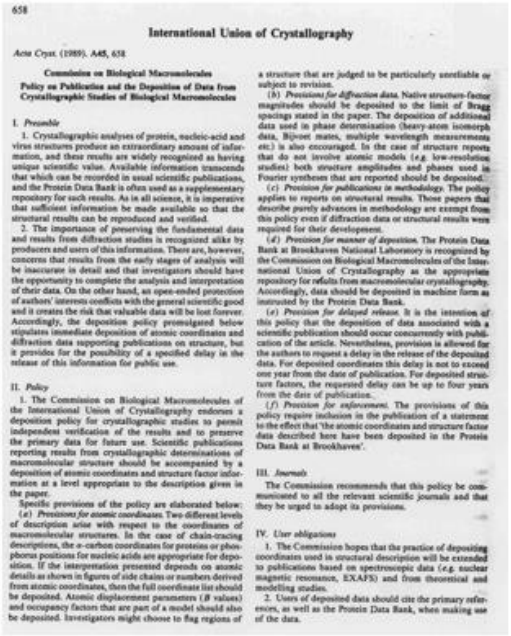
Ethics, Law & Policy – Community Driven Data Sharing

- Individual letters to editors of journals
- Committees
  - IUCr commission on Biological Macromolecules
  - ACA/USNCCr
  - Richards committee
- Funding agencies
- Articles in journals


Marvin Cassman


Fred Richards


Richard Dickerson


658

International Union of Crystallography

Acta Cryst. (1989). A45, 658

Commission on Biological Macromolecules

Policy on Publication and the Deposition of Data from Crystallographic Studies of Biological Macromolecules

I. *Preamble*

1. Crystallographic analyses of protein, nucleic-acid and virus structures produce an extraordinary amount of information, and these results are widely recognized as having unique scientific value. Available information transcends that which can be recorded in usual scientific publications, and the Protein Data Bank is often used as a supplementary repository for such results. As in all science, it is imperative that sufficient information be made available so that the structural results can be reproduced and verified.

2. The importance of preserving the fundamental data and results from diffraction studies is recognized alike by producers and users of this information. There are, however, concerns that results from the early stages of analysis will be inaccurate in detail and that investigators should have the opportunity to complete the analysis and interpretation of their data. On the other hand, an open-ended protection of authors' interests conflicts with the general scientific good and it creates the risk that valuable data will be lost forever. Accordingly, the deposition policy promulgated below stipulates immediate deposition of atomic coordinates and diffraction data supporting publications on structure, but it provides for the possibility of a specified delay in the release of this information for public use.

II. *Policy*

1. The Commission on Biological Macromolecules of the International Union of Crystallography endorses a deposition policy for crystallographic studies to permit independent verification of the results and to preserve the primary data for future use. Scientific publications reporting results from crystallographic determinations of macromolecular structure should be accompanied by a deposition of atomic coordinates and structure factor information at a level appropriate to the description given in the paper.

Specific provisions of the policy are elaborated below:

(*a*) *Provisions for atomic coordinates.* Two different levels of description arise with respect to the coordinates of macromolecular structures. In the case of chain-tracing descriptions, the α-carbon coordinates for proteins or phosphorus positions for nucleic acids are appropriate for deposition. If the interpretation presented depends on atomic details as shown in figures of side chains or numbers derived from atomic coordinates, then the full coordinate list should be deposited. Atomic displacement parameters (B values) and occupancy factors that are part of a model should also be deposited. Investigators might choose to flag regions of a structure that are judged to be particularly unreliable or subject to revision.

(*b*) *Provisions for diffraction data.* Native structure-factor magnitudes should be deposited to the limit of Bragg spacings stated in the paper. The deposition of additional data used in phase determination (heavy-atom isomorph data, Bijvoet mates, multiple wavelength measurements etc.) is also encouraged. In the case of structure reports that do not involve atomic models (e.g. low-resolution studies) both structure amplitudes and phases used in Fourier syntheses that are reported should be deposited.

(*c*) *Provision for publications in methodology.* The policy applies to reports on structural results. Those papers that describe purely advances in methodology are exempt from this policy even if diffraction data or structural results were required for their development.

(*d*) *Provision for manner of deposition.* The Protein Data Bank at Brookhaven National Laboratory is recognized by the Commission on Biological Macromolecules of the International Union of Crystallography as the appropriate repository for reflects from macromolecular crystallography. Accordingly, data should be deposited in machine form as instructed by the Protein Data Bank.

(*e*) *Provision for delayed release.* It is the intention of this policy that the deposition of data associated with a scientific publication should occur concurrently with publication of the article. Nevertheless, provision is allowed for the authors to request a delay in the release of the deposited data. For deposited coordinates this delay is not to exceed one year from the date of publication. For deposited structure factors, the requested delay can be up to four years from the date of publication.

(*f*) *Provision for enforcement.* The provisions of this policy require inclusion in the publication of a statement to the effect that 'the atomic coordinates and structure factor data described here have been deposited in the Protein Data Bank at Brookhaven'.

III. *Journals*

The Commission recommends that this policy be communicated to all the relevant scientific journals and that they be urged to adopt its provisions.

IV. *User obligations*

1. The Commission hopes that the practice of depositing coordinates used in structural description will be extended to publications based on spectroscopic data (*e.g.* nuclear magnetic resonance, EXAFS) and from theoretical and modelling studies.

2. Users of deposited data should cite the primary references, as well as the Protein Data Bank, when making use of the data.

Where Do We Go From Here As Data Scientists?

- Wikidata
  - Community curated knowledge graph
  - Broad coverage



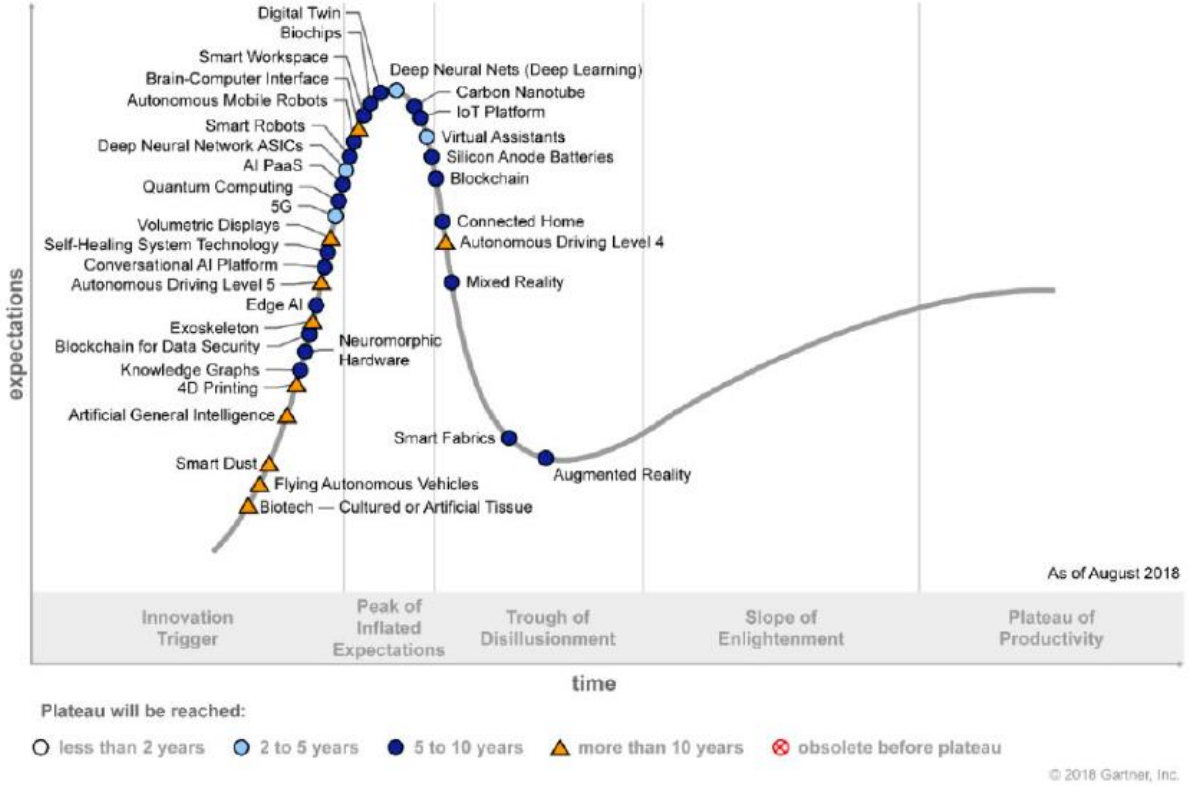
- Get on board with developments in schema.org, knowledge graphs, etc… as part of the rule rather than the exception
- Provide metadata and opinion for data we produce or use

Where Do You Go From Here?

- Follow the fourth paradigm - The data driven economy writ large will drive more interest in structured data
- There is the opportunity to contribute but also the opportunity to gain from a broader spectrum of FAIR data of different types
- Be patient…

Knowledge Graphs - Hype Technology?

metaphacts



Haas & Schmidt 2018
http://iswc2018.semanticweb.org/workshops-tutorials/#ekg

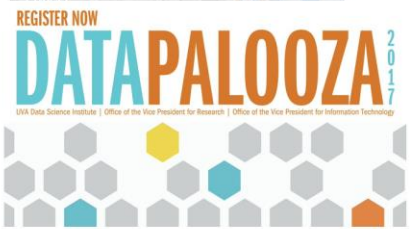
Acknowledgements

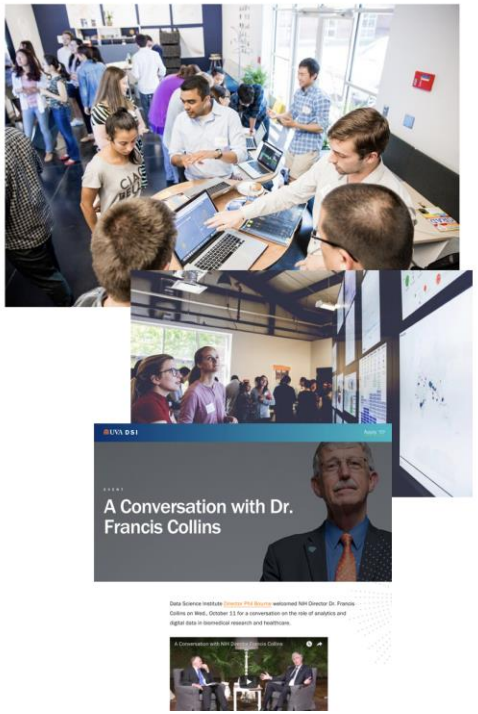


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DATA SCIENCE INSTITUTE

Working across the grounds to break down traditional silos



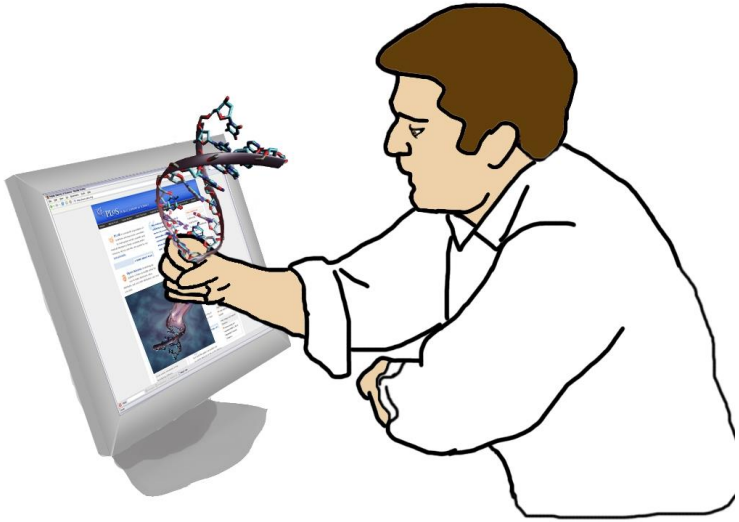




The 150 folks who have passed through my laboratory

https://docs.google.com/spreadsheets/d/1QZ48UaKcwDl_iFCvBmJsT03FK-bMchdfuIHe9Oxc-rw/edit#gid=0

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Thank You